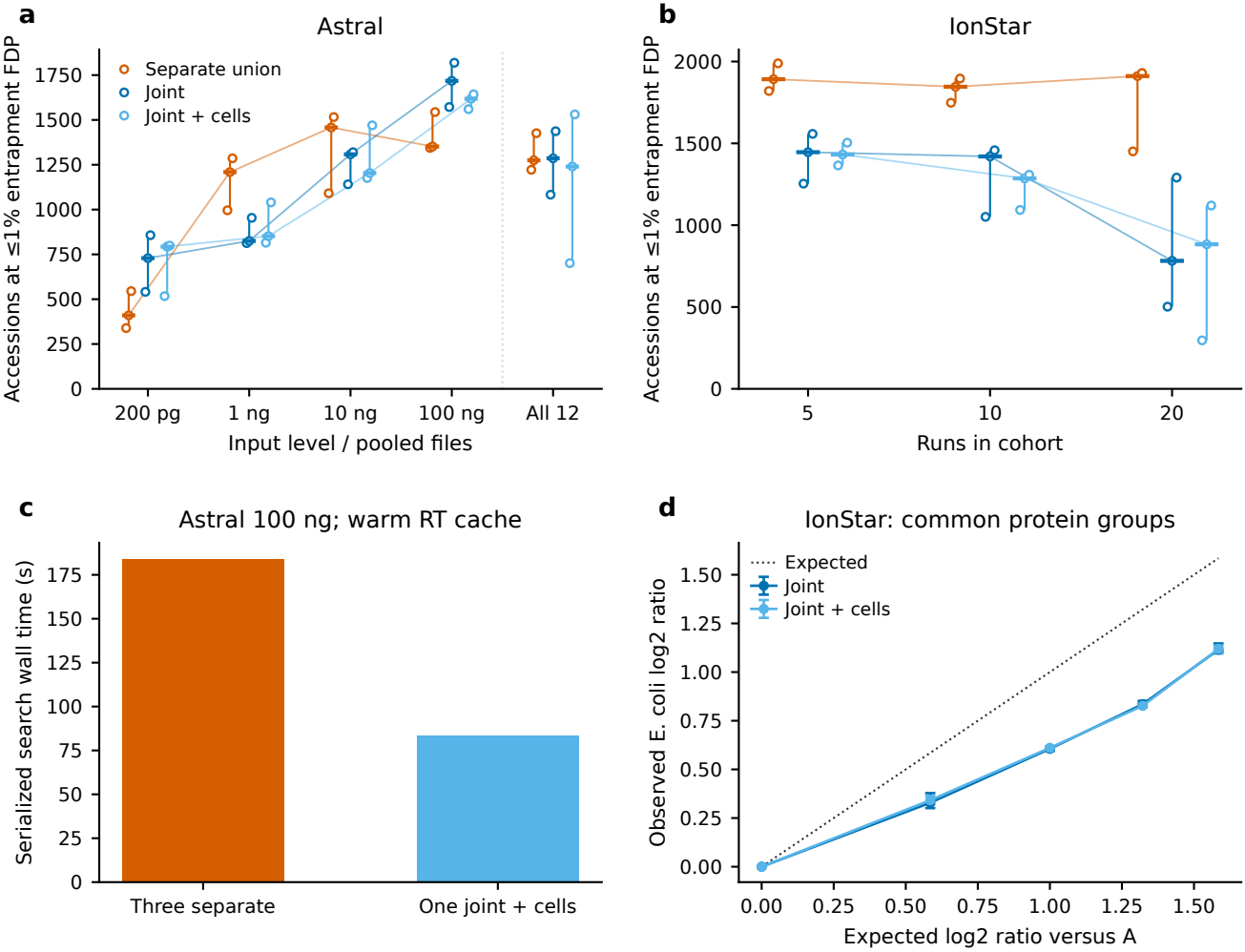

a
Astral
Separate union
Joint
Joint + cells
Accessions at ≤1% entrapment FDP
0
250
500
750
1000
1250
1500
1750
200 pg
1 ng
10 ng
100 ng
All 12
Input level / pooled files
b
IonStar
Accessions at ≤1% entrapment FDP
0
500
1000
1500
2000
5
10
20
Runs in cohort
c
Astral 100 ng; warm RT cache
Serialized search wall time (s)
0
25
50
75
100
125
150
175
Three separate
One joint + cells
d
IonStar: common protein groups
Expected
Joint
Joint + cells
Observed E. coli log2 ratio
0.00
0.25
0.50
0.75
1.00
1.25
1.50
Expected log2 ratio versus A